

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 2)

Let P be a point on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. Let the line passing through P and parallel to y -axis meet the circle $x^2 + y^2 = 9$ at point Q such that P and Q are on the same side of the x -axis. Then, the eccentricity of the locus of the point R on PQ such that $PR : RQ = 4 : 3$ as P moves on the ellipse, is :

(1) $\frac{11}{19}$

(2) $\frac{13}{21}$

(3) $\frac{\sqrt{139}}{23}$

(4) $\frac{\sqrt{13}}{7}$

Q2 - 2024 (27 Jan Shift 1)

The length of the chord of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$, whose mid point is $\left(1, \frac{2}{5}\right)$, is equal to :

(1) $\frac{\sqrt{1691}}{5}$

(2) $\frac{\sqrt{2009}}{5}$

(3) $\frac{\sqrt{1741}}{5}$

(4) $\frac{\sqrt{1541}}{5}$

Q3 - 2024 (29 Jan Shift 1)

If the points of intersection of two distinct conics $x^2 + y^2 = 4b$ and $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ lie on the curve $y^2 = 3x^2$, then $3\sqrt{3}$ times the area of the rectangle formed by the intersection points is _____

Q4 - 2024 (30 Jan Shift 1)

If the length of the minor axis of ellipse is equal to half of the distance between the foci, then the eccentricity of the ellipse is :

(1) $\frac{\sqrt{5}}{3}$

(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{1}{\sqrt{3}}$

(4) $\frac{2}{\sqrt{5}}$

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Questions with Answer Keys

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Q5 - 2024 (30 Jan Shift 2)

Let $A(\alpha, 0)$ and $B(0, \beta)$ be the points on the line $5x + 7y = 50$. Let the point P divide the line segment AB internally in the ratio $7 : 3$. Let $3x - 25 = 0$ be a directrix of the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the corresponding focus be S . If from S , the perpendicular on the x -axis passes through P , then the length of the latus rectum of E is equal to

- (1) $\frac{25}{3}$
- (2) $\frac{32}{9}$
- (3) $\frac{25}{9}$
- (4) $\frac{32}{5}$

Q6 - 2024 (31 Jan Shift 2)

Let P be a parabola with vertex $(2, 3)$ and directrix $2x + y = 6$. Let an ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$ of eccentricity $\frac{1}{\sqrt{2}}$ pass through the focus of the parabola P . Then the square of the length of the latus rectum of E , is

- (1) $\frac{385}{8}$
- (2) $\frac{347}{8}$
- (3) $\frac{512}{25}$
- (4) $\frac{656}{25}$

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Answer Key

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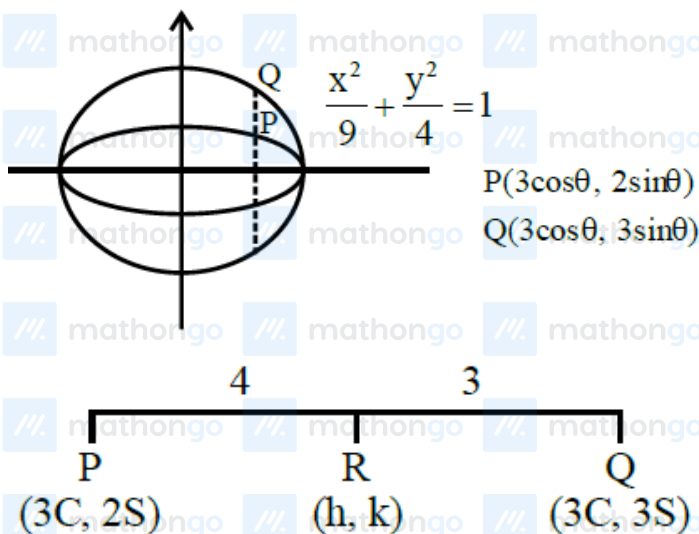
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Solutions

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Q1



$$h = 3 \cos \theta$$

$$k = \frac{18}{7} \sin \theta$$

$$\therefore \text{locus} = \frac{x^2}{9} + \frac{49y^2}{324} = 1$$

$$e = \sqrt{1 - \frac{324}{49 \times 9}} = \frac{\sqrt{117}}{21} = \frac{\sqrt{13}}{7}$$

Q2

Equation of chord with given middle point.

$$T = S_1$$

$$\frac{x}{25} + \frac{y}{40} = \frac{1}{25} + \frac{1}{100}$$

$$\frac{8x + 5y}{200} = \frac{8 + 2}{200}$$

$$y = \frac{10-8x}{5} \dots (i)$$

$$\frac{x^2}{25} + \frac{(10-8x)^2}{400} = 1 \quad (\text{put in original equation})$$

$$\frac{16x^2 + 100 + 64x^2 - 160x}{400} = 1$$

$$4x^2 - 8x - 15 = 0$$

$$x = \frac{8 \pm \sqrt{304}}{8}$$

$$x_1 = \frac{8 + \sqrt{304}}{8}; x_2 = \frac{8 - \sqrt{304}}{8}$$

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Solutions

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$$\text{Similarly, } y = \frac{10-18 \pm \sqrt{304}}{5} = \frac{2 \pm \sqrt{304}}{5}$$

$$y_1 = \frac{2-\sqrt{304}}{5}; y_2 = \frac{2+\sqrt{304}}{5}$$

$$\text{Distance} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$= \sqrt{\frac{4 \times 304}{64} + \frac{4 \times 304}{25}} = \frac{\sqrt{1691}}{5}$$

Q3

Putting $y^2 = 3x^2$ in both the conics

$$\text{We get } x^2 = b \text{ and } \frac{b}{16} + \frac{3}{b} = 1$$

$$\Rightarrow b = 4, 12 \quad (b = 4 \text{ is rejected because curves coincide})$$

$$\therefore b = 12$$

Hence points of intersection are

$$(\pm\sqrt{12}, \pm6) \Rightarrow \text{area of rectangle} = 432$$

Q4

$$2b = ae$$

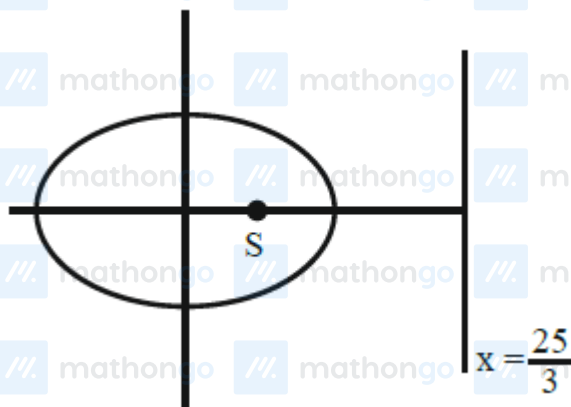
$$\frac{b}{a} = \frac{e}{2}$$

$$e = \sqrt{1 - \frac{e^2}{4}}$$

$$e = \frac{2}{\sqrt{5}}$$

Q5

$$\left. \begin{aligned} A &= (10, 0) \\ B &= \left(0, \frac{50}{7}\right) \end{aligned} \right\} P = (3, 5)$$



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Solutions

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$$ae = 3$$

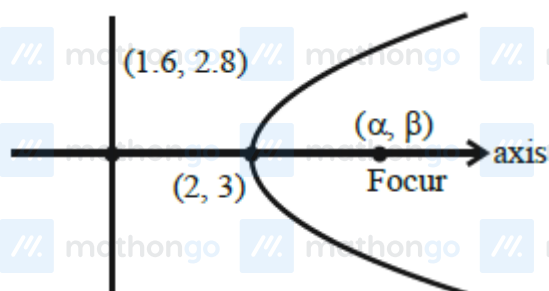
$$\frac{a}{e} = \frac{25}{3}$$

$$a = 5$$

$$b = 4$$

$$\text{Length of } LR = \frac{2b^2}{a} = \frac{32}{5}$$

Q6



$$\text{Slope of axis} = \frac{1}{2}$$

$$y - 3 = \frac{1}{2}(x - 2)$$

$$\Rightarrow 2y - 6 = x - 2$$

$$\Rightarrow 2y - x - 4 = 0$$

$$2x + y - 6 = 0$$

$$4x + 2y - 12 = 0$$

$$\alpha + 1.6 = 4 \Rightarrow \alpha = 2.4$$

$$\beta + 2.8 = 6 \Rightarrow \beta = 3.2$$

Ellipse passes through (2.4, 3.2)

$$\Rightarrow \frac{\left(\frac{24}{10}\right)^2}{a^2} + \frac{\left(\frac{32}{10}\right)^2}{b^2} = 1$$

$$\text{Also } 1 - \frac{b^2}{a^2} = \frac{1}{2} = \frac{b^2}{a^2} = \frac{1}{2}$$

$$\Rightarrow a^2 = 2b^2$$

$$\text{Put in (1)} \Rightarrow b^2 = \frac{328}{25}$$

$$\Rightarrow \left(\frac{2b^2}{a}\right)^2 = \frac{4b^2}{a^2} \times b^2 = 4 \times \frac{1}{2} \times \frac{328}{25} = \frac{656}{25}$$